SHARPE'S THEOREM ON LINEAR EQUATIONS, AND ITS UNIT FORMALISATION

Let R be a commutative ring and F an R-module. A system of m linear equations in n unknowns

$$\sum_{j=1}^{n} u_{ij}\, x_j = v_i \qquad (1 \le i \le m),$$

with $u_{ij} \in R$ and $v_i \in F$, to be solved in F, has at most one solution when the ideal $D_n(U)$ of n-minors of $U = (u_{ij})$ has grade at least 1 on F, and every solution makes the $(n+1)$-minors of $W = [U \mid v]$ vanish. Sharpe [4] proved that grade at least 2 gives the converse: if $\mathrm{Gr}(D_n(U); F) \ge 2$ and the $(n+1)$-minors of W vanish, the system has a solution. The hypothesis cannot be weakened to grade 1: over $\mathbb{Z}$ the system $2x = 3$, $4x = 6$ has vanishing 2-minors and $\mathrm{Gr}(2, 4) \ge 1$, but no solution. The theorem is Theorem B-7.5 of [1], where it is derived from a local-global principle for coverings in depth 2.

This note gives a short direct proof, in which the depth-2 hypothesis is used exactly once, on the Cramer numerators of the square subsystems, and describes its checked formalisation in Unit (`sag/depth2`). No localisation, prime ideal or Noetherian hypothesis is used.

1. Statement

As in [1, 2], depth is used in an explicit form, which for a finitely generated ideal is equivalent to the true grade of Hochster and Northcott [3].

Definition 1. Let $(d_\rho)_{\rho \in P}$ be a finite family of elements of R.

(1) $\mathrm{Gr}(d; F) \ge 1$ if $d_\rho x = 0$ for all ρ implies $x = 0$ ($x \in F$).
(2) $\mathrm{Gr}(d; F) \ge 2$ if moreover every family (y_ρ) of elements of F with $d_\rho y_{\rho'} = d_{\rho'} y_\rho$ for all ρ, ρ' is of the form $y_\rho = d_\rho e$ for some $e \in F$.

Under $\mathrm{Gr}(d; F) \ge 1$ the element e in (2) is unique. We index the minors by lists of rows. For $\rho : \{1, \dots, n\} \to \{1, \dots, m\}$, let U_ρ be the $n \times n$ matrix with rows $\rho(1), \dots, \rho(n)$ of U, and $\delta_\rho = \det U_\rho$. Rows may repeat and appear in any order; this gives 0 or $\pm$ a minor, so the family (δ_ρ) generates $D_n(U)$. For $\sigma : \{1, \dots, n+1\} \to \{1, \dots, m\}$ and $1 \le t \le n+1$, let $c_\sigma(t)$ be the determinant of the $(n+1) \times (n+1)$ matrix whose first n columns are those of U at the rows σ and whose last column is the unit vector e_t. The $(n+1)$-minor of W at σ is its expansion along the last column,

$$\mu_\sigma = \sum_{t=1}^{n+1} c_\sigma(t)\, v_{\sigma(t)} \in F;$$

for $F = R$ this is the determinant of the rows σ of W.

Theorem 2 (Sharpe)**.** *If $\mathrm{Gr}(\delta; F) \ge 2$ and $\mu_\sigma = 0$ for all σ, the system $Ux = v$ has a unique solution in F^n. Conversely, if it has a solution, then $\mu_\sigma = 0$ for all σ.*

2. Two determinant identities

For a square matrix A of size n, $A[k \leftarrow b]$ is A with column k replaced by b, and $\mathrm{adj}(A)_{kt} = \det A[k \leftarrow e_t]$. By linearity of det in one column, $(\mathrm{adj}(A)\, b)_k = \det A[k \leftarrow b]$.

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Lemma 3. $\operatorname{adj}(A)A = \det A{\cdot}I$ *and* $A\operatorname{adj}(A) = \det A{\cdot}I$. *Hence, for every vector* b, $\sum_k \det A[k \leftarrow b]\, a_k = \det A \cdot b$, *where* a_k *is the* k*-th column of* A.

Proof. The first identity is column linearity together with the vanishing of a determinant with a repeated column. For the second, it suffices to show $\operatorname{adj}(A^T) = \operatorname{adj}(A)^T$, i.e. $\det A^T[k \leftarrow e_i] = \det A[i \leftarrow e_k]$. Let Z be A with column i replaced by e_k and the other entries of row k replaced by 0. Then $A[i \leftarrow e_k]E = Z$ for the elementary matrix $E = I + e_i w^T$ with $w_l = -A_{kl}$ for $l \neq i$ and $w_i = 0$, so $\det A[i \leftarrow e_k] = \det Z$. The same construction for A^T, k and i gives Z^T, so $\det A^T[k \leftarrow e_i] = \det Z^T = \det Z$. □

Lemma 4. *Let* ρ *be a list of* n *rows,* r *a row and* $\sigma = (\rho, r)$. *For every* $v \in R^{n+1}$, *writing* $v' = (v_1, \ldots, v_n)$ *and* N_v *for the matrix with the columns of* U *at the rows* σ *and last column* v,

$$\det N_v + \textstyle\sum_{j=1}^{n} u_{rj} \det U_\rho[j \leftarrow v'] = \delta_\rho\, v_{n+1}.$$

Proof. Let $N = N_{e_{n+1}}$. The last row of N is $(u_{r1}, \ldots, u_{rn}, 1)$ and its last column is e_{n+1}, so $\det N = \delta_\rho$. Likewise $N[j \leftarrow v]$ has last column e_{n+1} and top left block $U_\rho[j \leftarrow v']$, so its determinant is $\det U_\rho[j \leftarrow v']$; and $N[n+1 \leftarrow v] = N_v$. The last coordinate of the identity of Lemma 3 for N and $b = v$ is the claim. □

3. Proof of the theorem

Let $X_{\rho,k} = \sum_t \operatorname{adj}(U_\rho)_{kt}\, v_{\rho(t)} \in F$, Cramer's numerators for the square subsystem of rows ρ, and write $(Ux)_r = \sum_j u_{rj} x_j$ for $x \in F^n$.

The rows. Applying Lemma 4 to $v = e_t$, multiplying by $v_{\sigma(t)}$ and summing over t gives, for all ρ and r,

$$\mu_{(\rho,r)} + (UX_\rho)_r = \delta_\rho\, v_r. \tag{1}$$

On the left, the term $t = n+1$ of the second sum vanishes, since $e'_{n+1} = 0$; the terms $t \leq n$ give X_ρ.

Uniqueness. For $x \in F^n$, $\sum_t \operatorname{adj}(U_\rho)_{kt}(Ux)_{\rho(t)} = (\operatorname{adj}(U_\rho)U_\rho\, x)_k = \delta_\rho x_k$ by Lemma 3. So if $Ux = Uy$, then $\delta_\rho x_k = \delta_\rho y_k$ for all ρ, and $x = y$ as soon as $\operatorname{Gr}(\delta; F) \geq 1$.

Existence. Assume all μ_σ vanish, so that $UX_\rho = \delta_\rho v$ for all ρ by (1). Then $U(\delta_{\rho'} X_\rho) = \delta_{\rho'}\delta_\rho v = U(\delta_\rho X_{\rho'})$, and by uniqueness $\delta_{\rho'} X_{\rho,k} = \delta_\rho X_{\rho',k}$ for every k. So for each k the family $(X_{\rho,k})_\rho$ is proportional to $(\delta_\rho)_\rho$, and $\operatorname{Gr}(\delta; F) \geq 2$ gives $x_k \in F$ with $X_{\rho,k} = \delta_\rho x_k$ for all ρ. Then $\delta_\rho (Ux)_r = (UX_\rho)_r = \delta_\rho v_r$ for all ρ, hence $Ux = v$.

Converse. If $Ux = v$, then $\mu_\sigma = \sum_j x_j \sum_t c_\sigma(t) u_{\sigma(t)j}$, and the inner sum is the determinant of the rows σ of W with its last column replaced by column j of U: a repeated column, so 0. □

Remark 5. In [1] the existence is obtained by solving the system over each $R[1/\delta_\rho]$ by Cramer's rule and gluing the local solutions with the local-global principle B-7.4 for coverings in depth 2; Sharpe first treats an explicit regular sequence of length 2 in $D_n(U)$ and then passes to a polynomial extension. Here the local solutions are replaced by the numerators X_ρ, whose compatibility is an identity (uniqueness), and depth 2 is applied to them directly. The determinant identities that Sharpe establishes are Lemmas 3 and 4.

4. The formalisation

The theorem is checked in Unit, in `sag/depth2`, with univalence and replacement as parameters as in the rest of `sag` (`./segc sag/depth2 all`: 171 modules, about 2.6 s; every file of the development under 0.05 s). Matrices are lists of columns, as in `sag/exterior`, where

$\det$, $\mathrm{adj}(A)A = \det A \cdot I$, $\det A^T = \det A$ and $\det(AB) = \det A \det B$ are proved. Depth is the explicit form above (`Depth`); for a family indexed by $\{1, \dots, N\}$ it coincides with the Koszul grade of `sag/koszul` (`DepthKoszul`). The ring is the module `Modules.regular`, so the case $F = R$ is an instance; it is also proved separately.

`Depth`	$\mathrm{Gr} \geq 1$, $\mathrm{Gr} \geq 2$, uniqueness of the multiple
`Cofactor`	Lemma 3 (`symmetry`, `adj-right`, `cramer`), via `WiebeMatrices.emat`
`SharpeRows`	Lemma 4 (`Row`), via `WiebeMatrices.BlockDet`
`Sharpe`	Theorem 2 for $F = R$ (`System.theorem`, `sol-prop`, `necessary`)
`SharpeModule`	finite sums in a module, (1) (`RowF`), Theorem 2 (`System.theorem`, `sol-prop`, `necessary`)

The same directory contains the other results of [1, B-7] in depth 2: the local-global principles for divisibility and strong gcds (B-7.1), for depth 2 itself (B-7.2 with $k = 2$), and the covering principle (B-7.4), for localisations at single elements.

References

[1] T. Coquand, H. Lombardi, *Résolutions libres finies. Méthodes constructives*, Calvage & Mounet, 2024.
[2] T. Coquand, C. Quitté, *Constructive finite free resolutions*, manuscripta math. 137 (2012), 331–345.
[3] D. G. Northcott, *Finite Free Resolutions*, Cambridge Tracts in Mathematics 71, Cambridge University Press, 1976.
[4] D. W. Sharpe, *Grade and the theory of linear equations*, Linear Algebra Appl. 18 (1977), 25–32.